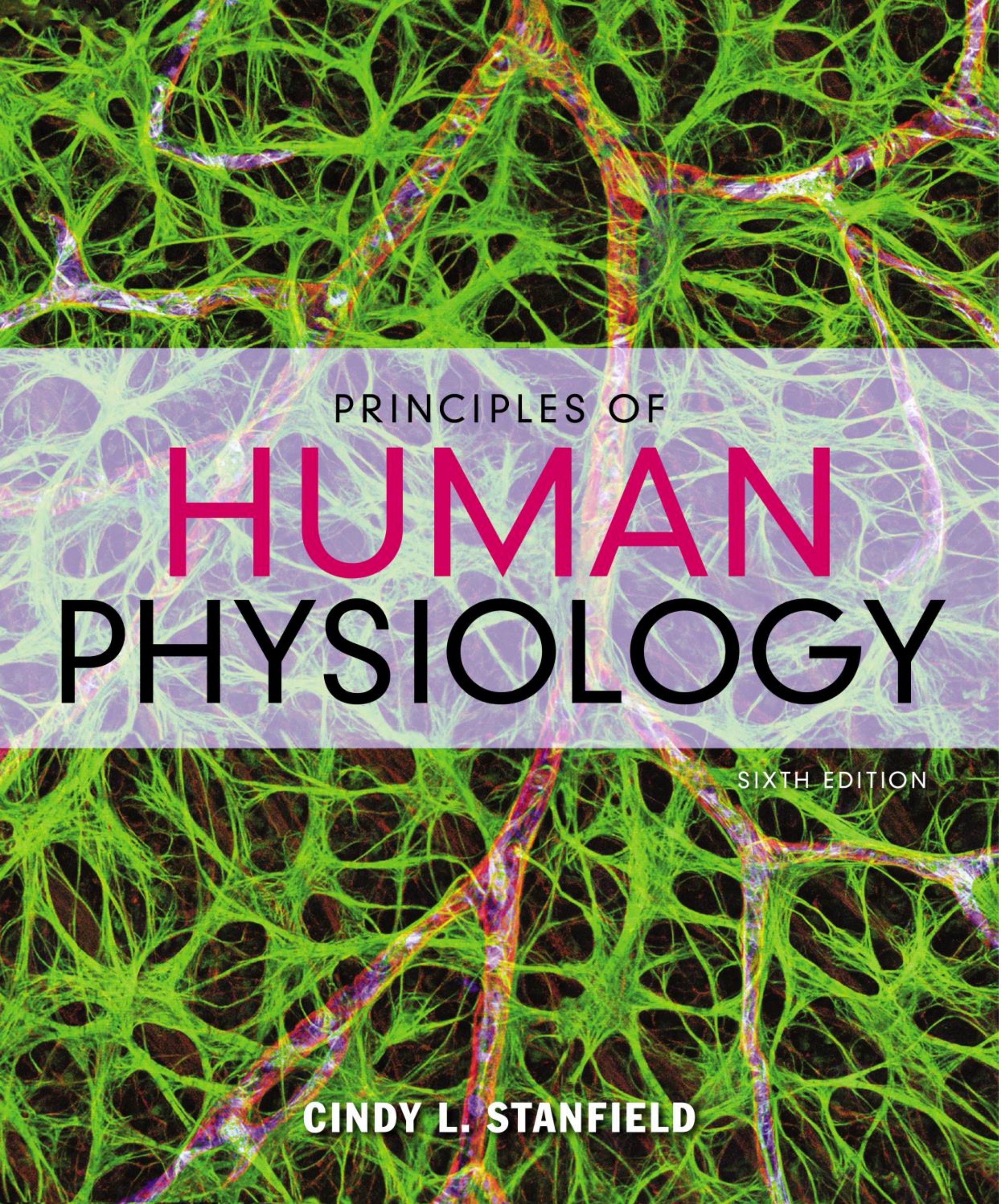


PowerPoint® Lecture Presentation

CHAPTER 23b

The Immune System

Chapter Outline

23.4 Humoral Immunity

- Role of B lymphocytes in antibody production
  - B lymphocyte exposure to antigen triggers clonal selection
    - Memory B cells + plasma cells
  - Plasma cells
    - Life span: 4–7 days
    - Secrete 2000 antibodies specific for antigen per second
    - Antibodies circulate several weeks, binding to and marking antigen for destruction
      - Phagocytosis
      - Complement-mediated lysis

Role of B Lymphocytes in Antibody Production

- Sometimes proliferation and differentiation of B cells in response to antigen depends on helper T cells
  - Antigens: T-dependent antigens
  - Helper T cells respond to specific antigen
    - Secrete IL-2 (among other cytokines)
- IL-2 + T-dependent antigens → activate B cells

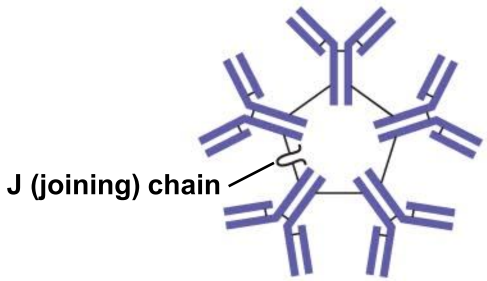
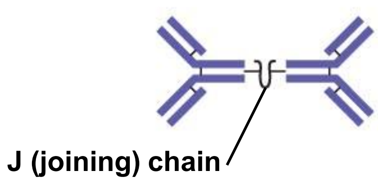
Role of B Lymphocytes in Antibody Production

- B cells can proliferate and differentiate into plasma cells (but not memory cells) in the absence of T cells
- Antigens: polysaccharides on bacterial surfaces
- Response weaker than helper T cell–dependent response
- No memory for subsequent exposures

Antibody Function in Humoral Immunity

- Antibody functions
  - Binds to specific antigen
  - Aids in inactivation or destruction of antigen
- Antibody: antigen-binding site and tail
- Antigen-binding site recognizes and binds antigen
- Tail is involved in inactivation or destruction of antigen

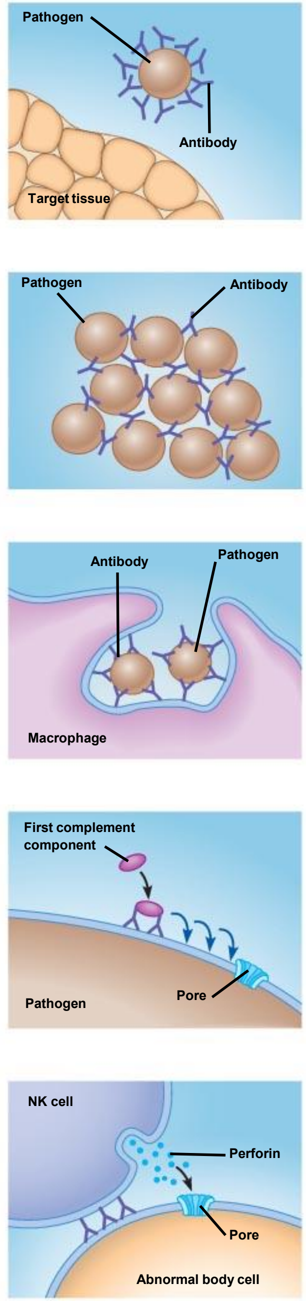
Figure 23.15 The five major classes of antibodies.

Class of antibody IgM	Structure	Features	Roles in antigen disposal
IgM		The most common class of antibody produced in the primary response to antigen	Neutralizes antigen Agglutinates antigen Activates complement
IgD		Important as an antigen receptor on B cells	Neutralizes antigen Agglutinates antigen
IgG		The most common class of antibody in the blood, and the major class of antibody produced in secondary responses. Crosses the placenta, so it is important in fetal and newborn immunity	Neutralizes antigen Agglutinates antigen Activates complement Opsonizes antigen Enhances NK cell activity
IgE		Involved in allergies such as hay fever	Neutralizes antigen Agglutinates antigen Binds to mast cells and basophils, causing them to release histamine
IgA		Crosses epithelial cells, so is present on mucosal surfaces and in breast milk; is important in immunity in newborns	Neutralizes antigen Agglutinates antigen

Antibody Function in Humoral Immunity

- Antibody actions
  - All classes: neutralization and agglutination
  - IgM and IgG: activate complement
  - IgG: opsonization
  - IgE: histamine release from mast cells and basophils
  - IgG: NK cells
    - NK cells have receptors for the antibody tail
    - Antibodies mark cells for destruction
    - NK cells produce pores in the membranes of cells, causing lysis

Figure 23.16 Antibody-mediated mechanisms of antigen disposal.



(a) Neutralization: Antibodies block the activity of a pathogen.

(b) Agglutination: Multiple pathogens are aggregated by antibody molecules.

(c) Opsonization: Pathogens bound by antibodies are more efficiently engulfed by phagocytes.

(d) Complement activation: Antibodies bound to pathogens activate the complement cascade, resulting in lysis of the cell.

(e) Enhanced NK cell activity: Abnormal body cells that are bound by antibodies are recognized by NK cells and are subsequently lysed.

23.5 Cell-Mediated Immunity

- Roles of T lymphocytes in cell-mediated immunity
  - Helper T cells
    - Secrete cytokines that enhance activity of B cells and other T cells
    - Enhance activity of macrophages and NK cells
  - Cytotoxic T cells
    - Kill virus-infected cells, abnormal cells, and bacteria
    - Suppressor T cells
      - Secrete cytokines that suppress activity of B cells and other T cells

Roles of T Lymphocytes in Cell-Mediated Immunity

- Major histocompatibility complex (MHC) molecules
  - T cells have antigen receptors
    - Only recognize antigen when it is associated with MHC molecules
  - MHC molecules bind antigen within the cell and transport antigen to the surface, where it can be recognized by T cells: antigen presentation

Roles of T Lymphocytes in Cell-Mediated Immunity

- MHC marks body cells as "self"
  - Two classes of MHC molecules
    - Class I MHC: surface of all nucleated cells
    - Class II MHC: surface of macrophages, activated B cells, activated T cells, and thymus cells
  - MHC molecules unique to individual person: human leukocyte antigen (HLA)
    - Tissue type
    - Marks person as "self"
    - Responsible for tissue or organ rejection—stimulates immune response to foreign tissue

Roles of T Lymphocytes in Cell-Mediated Immunity

- Final maturation of T cells depends on contact with MHC molecules
- Bind with Type I MHC → cytotoxic T (CD8) cells
- Bind with Type II MHC → helper T (CD4) cells

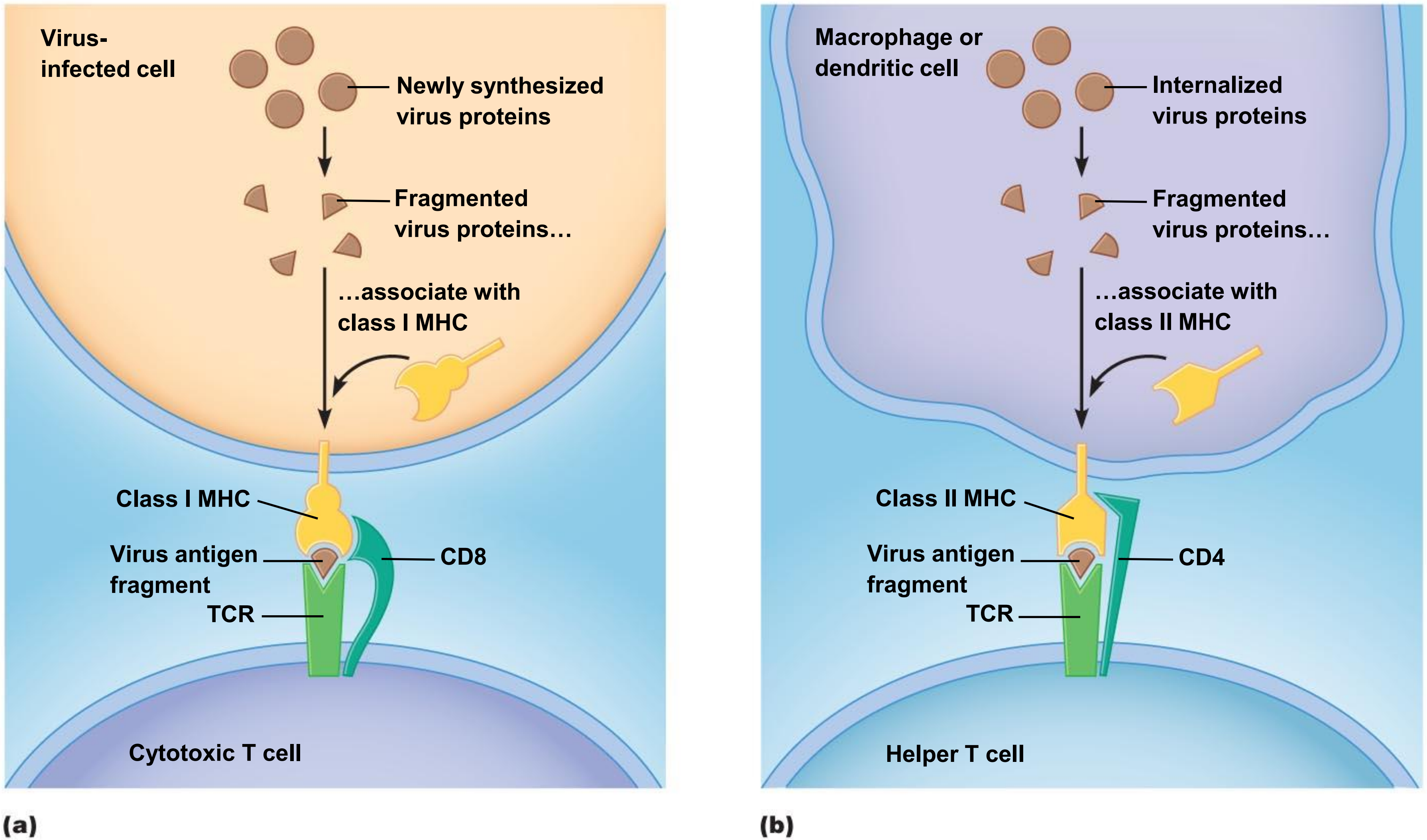
Roles of T Lymphocytes in Cell-Mediated Immunity

- Role of MHC molecules in antigen presentation
  - MHC molecules have binding sites for antigen
  - MHC + antigen → cell surface (presentation)
  - Type I MHC
    - Bind antigens in infected cells or tumor cells (which can be any nucleated cell)
  - Type II MHC
    - Bind antigens taken into cell by phagocytosis or receptor-mediated endocytosis (which can occur only in cells that perform these functions—B cells, T cells, macrophages)

Roles of T Lymphocytes in Cell-Mediated Immunity

- Recognition by T cells
  - Helper T cells
    - CD4: binds class II MHC; enhances macrophage–T cell interaction
  - Cytotoxic T cells
    - CD8: binds class I MHC; enhances abnormal cell–T cell interaction

Figure 23.17 Presentation of antigens to T cells by major histocompatibility complex (MHC) molecules.



Helper T Cell Activation

- Helper T cells coordinate immune response
- Events of activation
  - Bind to class II MHC–foreign antigen complex on surface of macrophages and B cells
  - Macrophages and B cells secrete IL-1, which induces proliferation and differentiation of helper T cells
- Helper T cell differentiation
  - Secretory cells (effector cells) and memory cells
  - Secretory cells: secrete cytokines that stimulate and regulate immune response

TABLE 23.1 Selected Cytokines Secreted by Activated Helper T Cells

Cytokine	Target cells	Effects on target cells
Interleukin-2	Helper T cells	Stimulates proliferation
	Cytotoxic T cells	Stimulates proliferation
	B cells	Stimulates proliferation and plasma cell development
	NK cells	Enhances activity
Interleukin-4	B cells	Stimulates proliferation and plasma cell development; induces plasma cells to secrete IgE and IgG; increases numbers of class II MHC molecules
	Helper T cells	Stimulates proliferation
	Macrophages	Increases numbers of class II MHC molecules; enhances phagocytosis
	Mast cells	Stimulates proliferation
Interleukin-5	B cells	Stimulates proliferation; induces plasma cells to secrete IgA
	Hematopoietic stem cells	Induces proliferation and development of eosinophils
Interleukin-10	Macrophages	Inhibits cytokine production (helps down-regulate the immune response)
Interferon-γ	Multiple cell types	Confers resistance to viruses
	Macrophages	Enhances phagocytosis
	B cells	Enhances antibody production
	Cancer cells	Inhibits proliferation
	Cytotoxic T cells and NK cells	Enhances killing capacity of cytotoxic T cells and NK cells

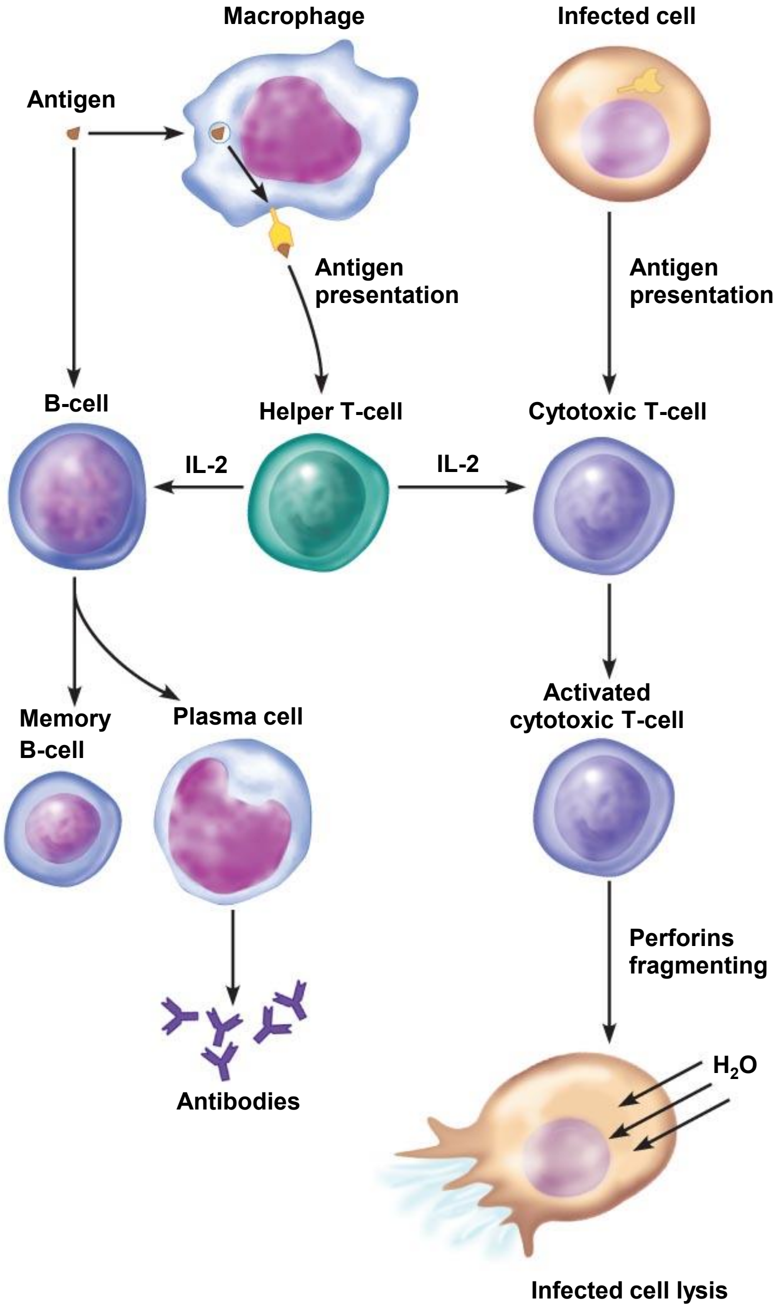
Cytotoxic T Cell Activation: The Destruction of Virus-Infected Cells and Tumor Cells

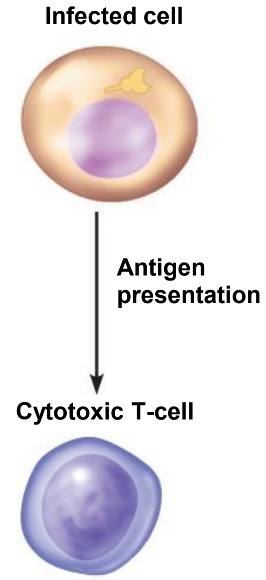
- Cell binds to class I MHC–foreign antigen complex on surface of infected cell or tumor cell
- Helper T cell secretes IL-2, which signals activation of cytotoxic T cell
- Cell secretes perforins
  - Form pores in the membranes of infected cells
  - Lysis
- Cell secretes fragmentins
  - Enter infected cells through perforin-induced pores
  - Trigger apoptosis

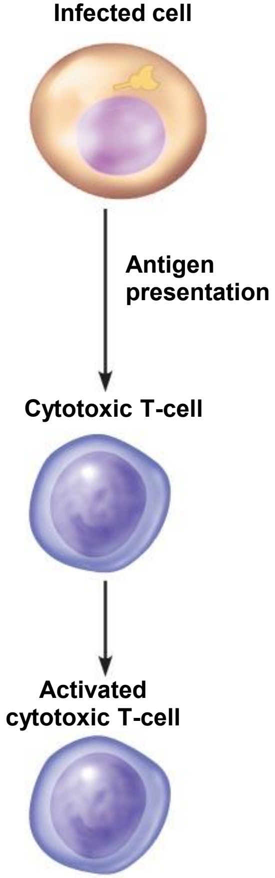
Cytotoxic T Cell Activation: The Destruction of Virus-Infected Cells and Tumor Cells

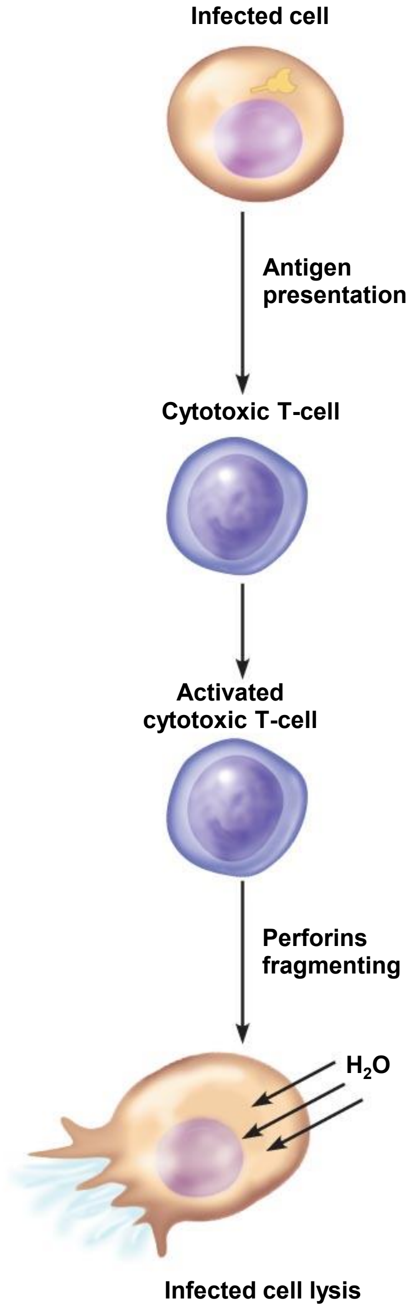
- Tumor cells → tumor antigens
- Class I MHC molecules bring antigens to surface and present them to cytotoxic T cells
- Cytotoxic T cells destroy tumor cells
- Some cancers and viruses inhibit production of class I MHC molecules, thereby protecting them from the immune system

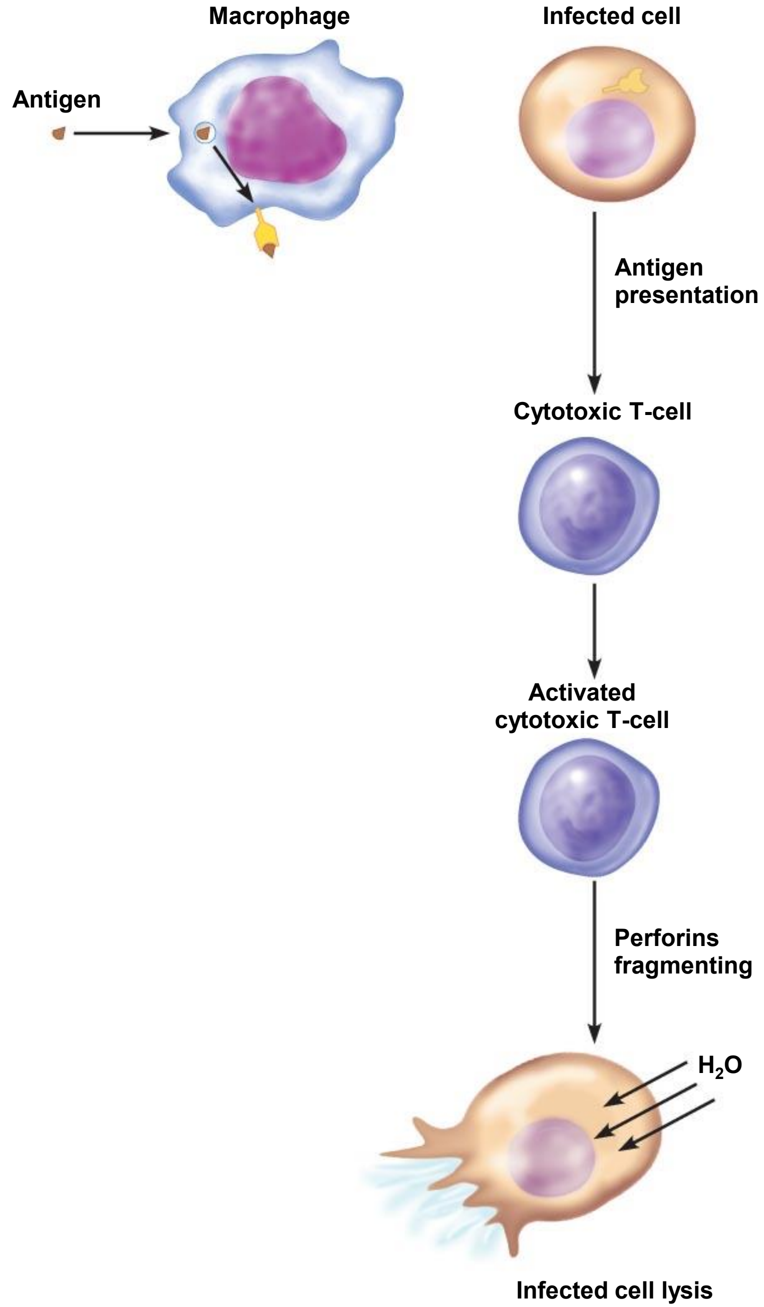
Figure 23.18 The role of helper T cells in activation of B cells and cytotoxic T cells.

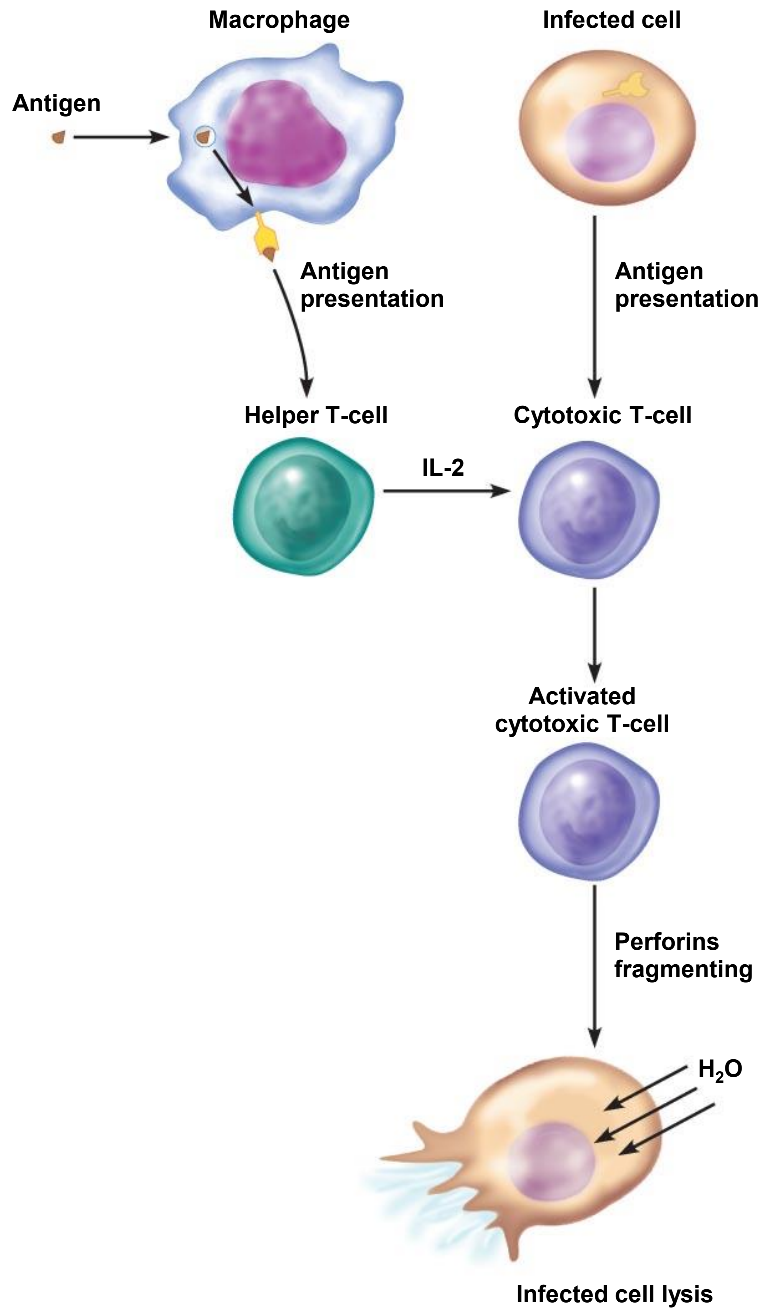



Infected cell
Antigen
presentation
Cytotoxic T-cell


Infected cell
Antigen
presentation
Cytotoxic T-cell
Activated
cytotoxic T-cell


Infected cell
Antigen
presentation
Cytotoxic T-cell
Activated
cytotoxic T-cell
Perforins
fragmenting
H_2O
Infected cell lysis


Macrophage
Antigen
Infected cell
Antigen
presentation
Cytotoxic T-cell
Activated
cytotoxic T-cell
Perforins
fragmenting
H_2O
Infected cell lysis


Macrophage
Infected cell
Antigen
Antigen presentation
Antigen presentation
Helper T-cell
Cytotoxic T-cell
IL-2
Activated cytotoxic T-cell
Perforins fragmenting
H_2O
Infected cell lysis

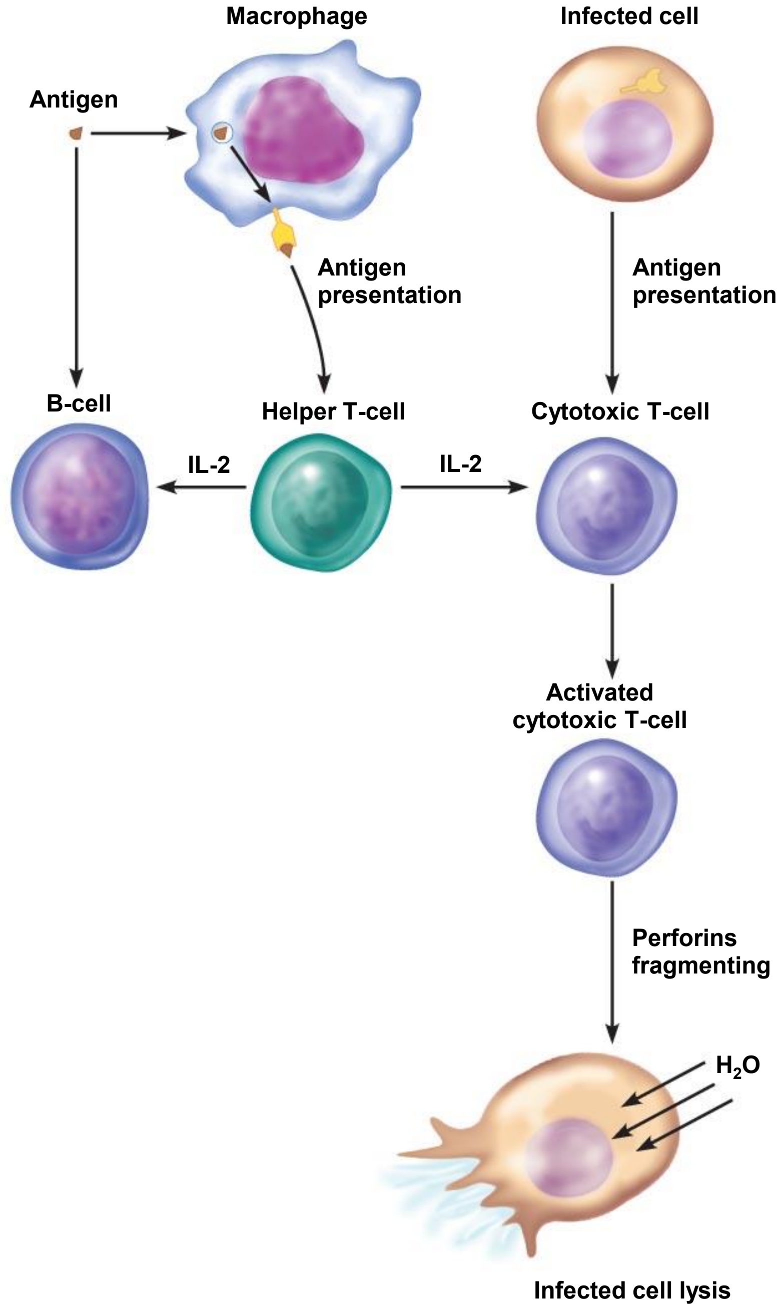
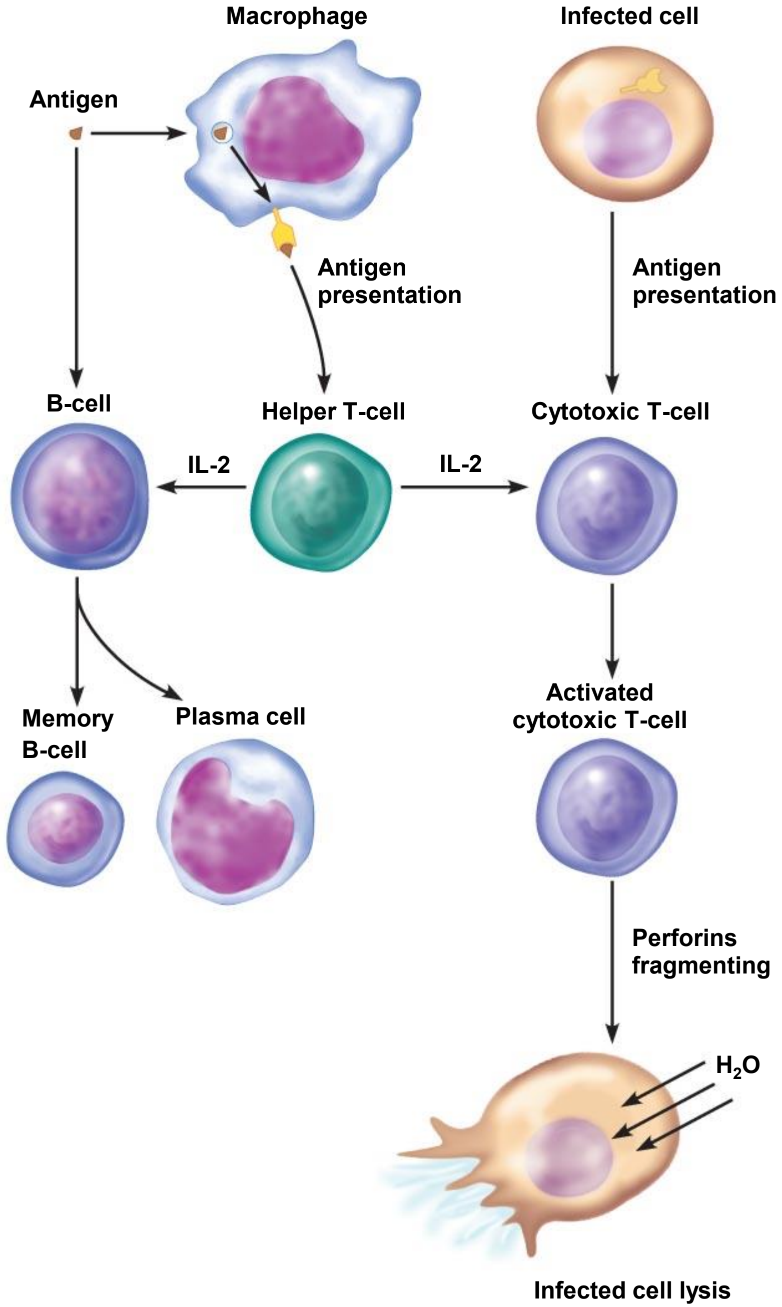
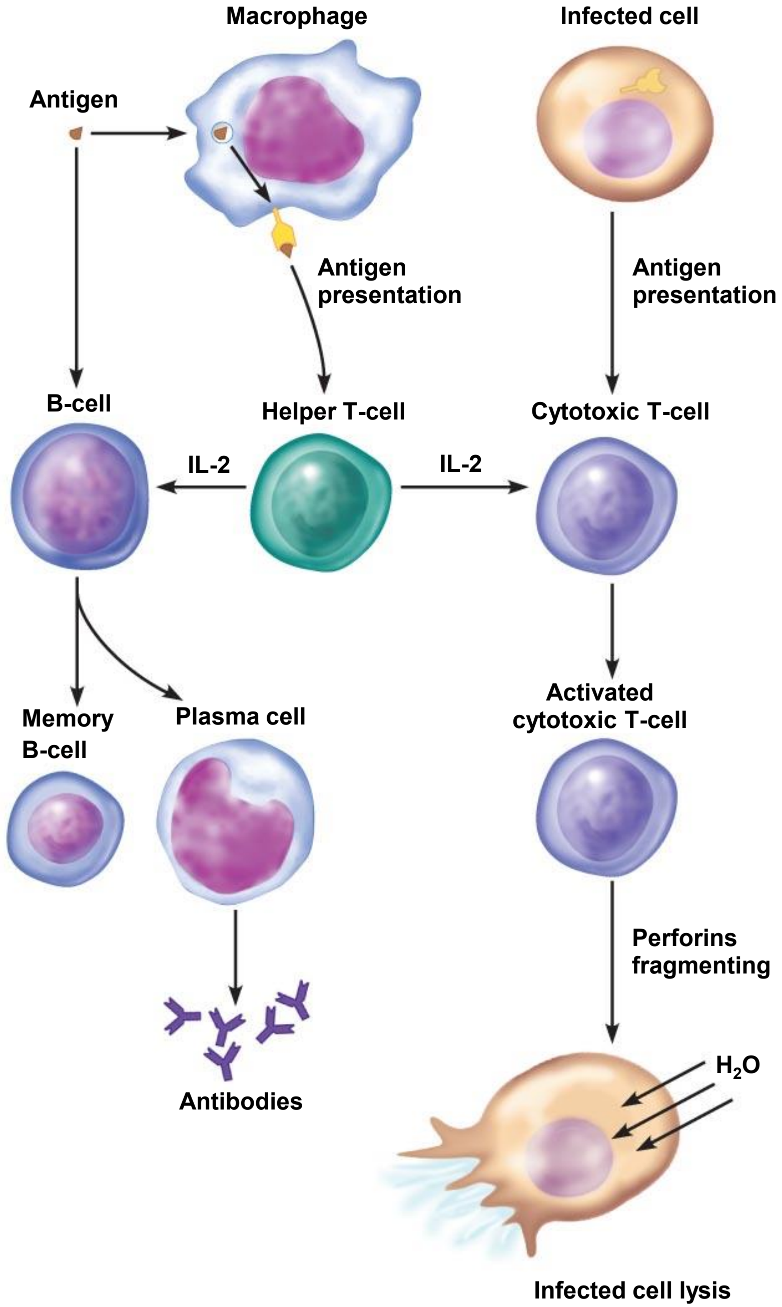

Macrophage
Infected cell
Antigen
Antigen
presentation
Antigen
presentation
B-cell
Helper T-cell
Cytotoxic T-cell
IL-2
IL-2
Activated
cytotoxic T-cell
Perforins
fragmenting
H_2O
Infected cell lysis

Figure 23.18 The role of helper T cells in activation of B cells and cytotoxic T cells.





Macrophage
Infected cell
Antigen
Antigen presentation
Antigen presentation
B-cell
Helper T-cell
Cytotoxic T-cell
IL-2
IL-2
Memory B-cell
Plasma cell
Activated cytotoxic T-cell
Perforins fragmenting
H_2O
Antibodies
Infected cell lysis

23.6 Immune Responses in Health and Disease

- Generating immunity: immunization
  - Vaccine: introduction of microorganism or its antigens in a form not expected to cause disease
  - Induces immune response, including production of memory cells
- Active immunity
  - Immune response to vaccine or pathogen in individual gives immunity
- Passive immunity
  - Administration of ready-made antibodies
  - No memory cells, so no long-term immunity

Figure 23.19 Acquisition of long-term immunity through vaccination (immunization).

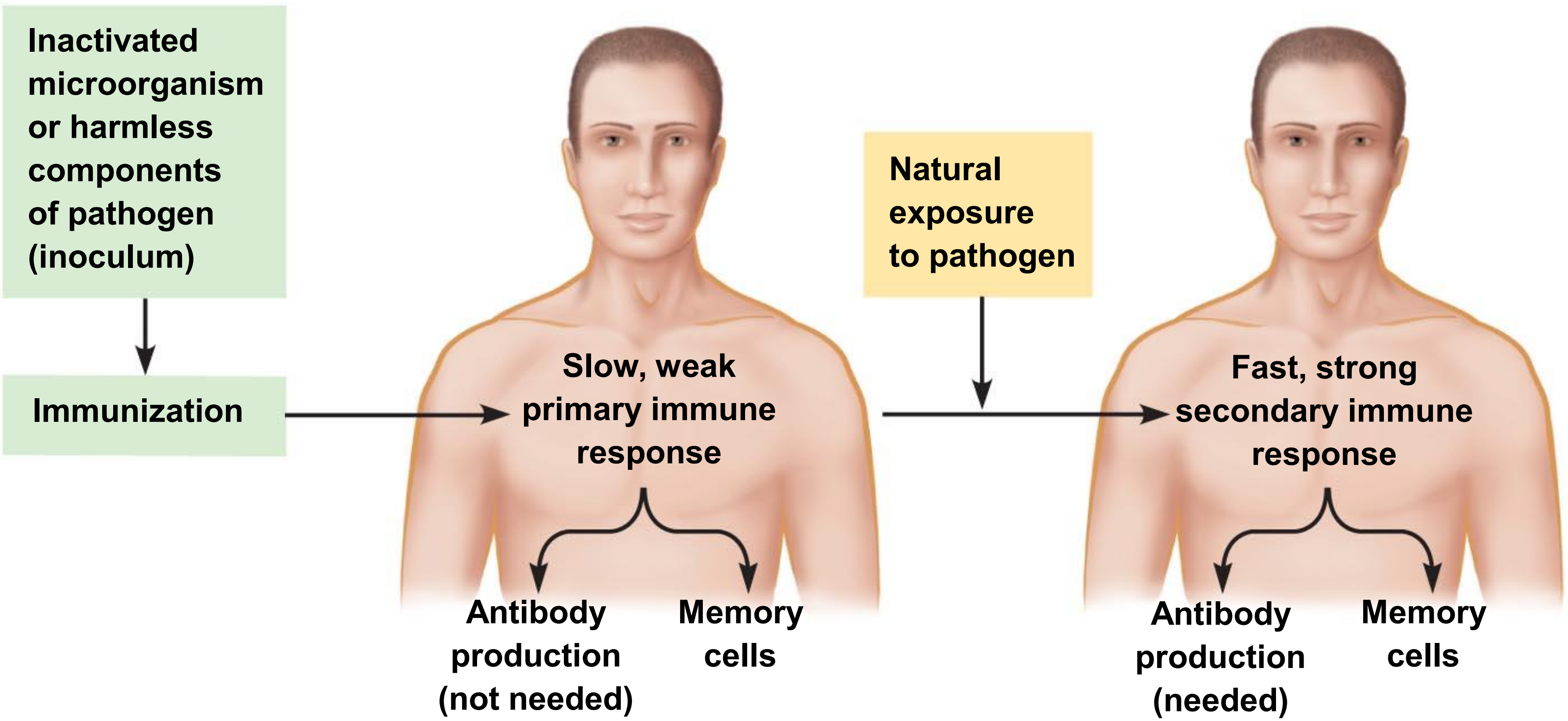


TABLE 23.2 Recommended 2012 Vaccinations by the Centers for Disease Control and Prevention

Vaccine	Regimen
Hepatitis B (HepB)	Sequence of 3 shots at birth, 1–2 months, and 6–18 months
Rotavirus (RV)	Sequence of 3 shots at 2, 4, and 6 months
Diphtheria, tetanus, pertussis (DTaP)	Sequence of 6 shots at 2 months, 4 months, 6 months, 15–18 months, 4–6 years, and 11–12 years, then tetanus every 10 years
Haemophilus influenzae type b (Hib)	Sequence of 4 shots at 2, 4, 6, and 12–15 months
Pneumococcal	Sequence of 4 shots at 2, 4, 6, and 12–15 months
Inactivated polio virus	Sequence of 4 shots at 2 months, 4 months, 6–18 months, and 4–6 years
Influenza	Annually starting at 6 months
Measles, mumps, and rubella (MMR)	Two shots at 12–15 months and 4–6 years
Varicella (chicken pox)	Two shots at 12–15 months and 4–6 years
Hepatitis A (HepA)	Two shots at 12–24 months and then 6–18 months after the first short
Meningococcal (MCV)	One shot at 11–12 years and a booster at 16 years (can be used in at-risk infants)

Generating Immunity: Immunization

- Passive immunity from mother to fetus or baby
  - IgG passes to fetus via the placenta
  - IgA passes to baby in breast milk

Role of Immune System in Transfusion and Transplantation

- Immune system recognizes difference between self and non-self
- Need to match donor to recipient closely
- Must suppress immune system in recipient

Role of Immune System in Transfusion and Transplantation

- Blood group compatibility
  - Blood types: A, B, AB, O
    - Due to antigens on surface of RBCs
  - Type A → A antigens, anti-B antibodies
  - Type B → B antigens, anti-A antibodies
  - Type AB → A and B antigens, no A or B antibodies
  - Type O → no antigens, anti-A and anti-B antibodies
  - Universal donor: O
  - Universal recipient: AB

Role of Immune System in Transfusion and Transplantation

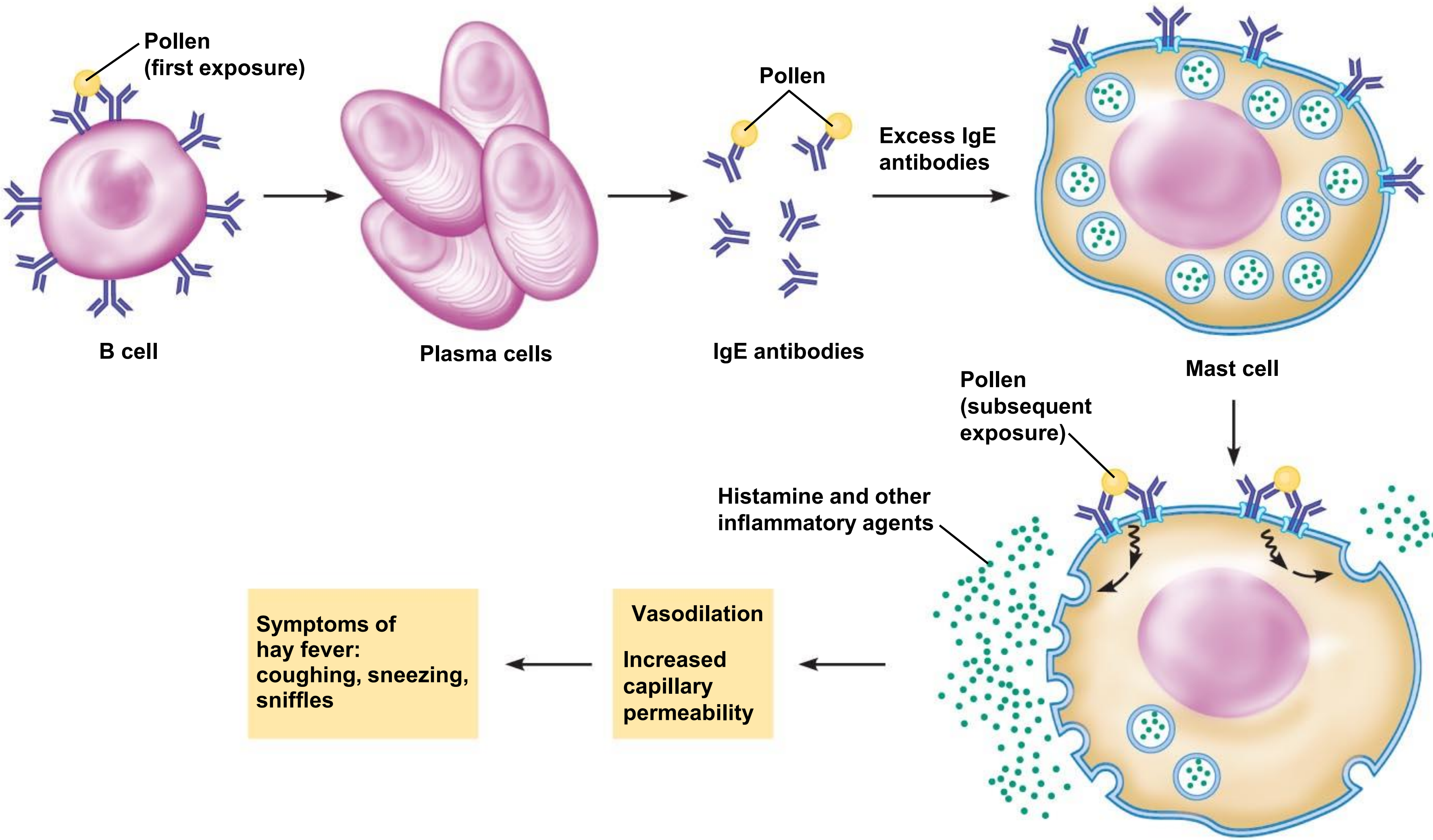
- Tissue grafts and organ transplantation
  - HLA molecules (MHC) stimulate rejection by inducing an immune response
  - Tissue typing: matches HLA molecules between donor and recipient
  - Must suppress immune response of recipient
  - Cyclosporin A and FK506 inhibit IL-2

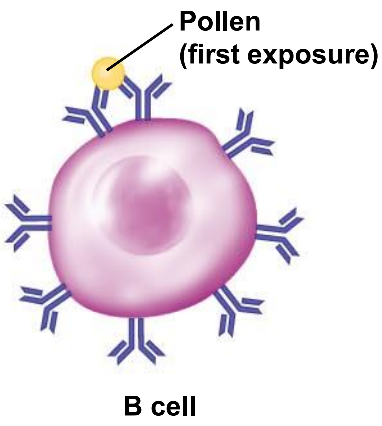
Role of Immune System in Transfusion and Transplantation

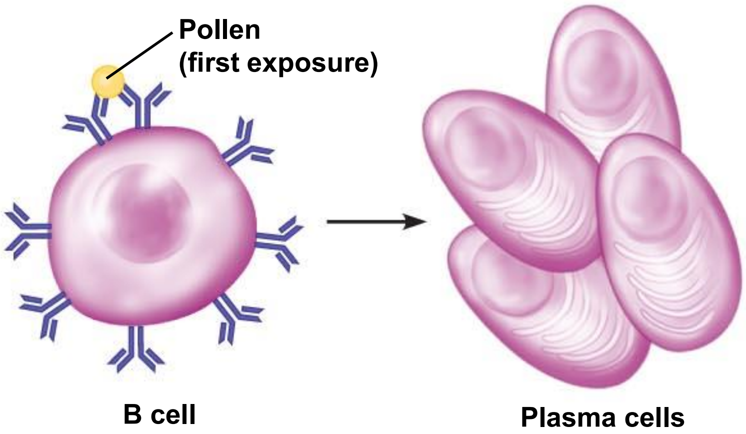
- Bone marrow transplants: some are more successful than others
- Radiation treatment destroys bone marrow cells, inactivating immune system
- Donated marrow may mount an immune response against recipient: graft versus host reaction
- Necessary to match donor and recipient HLA molecules

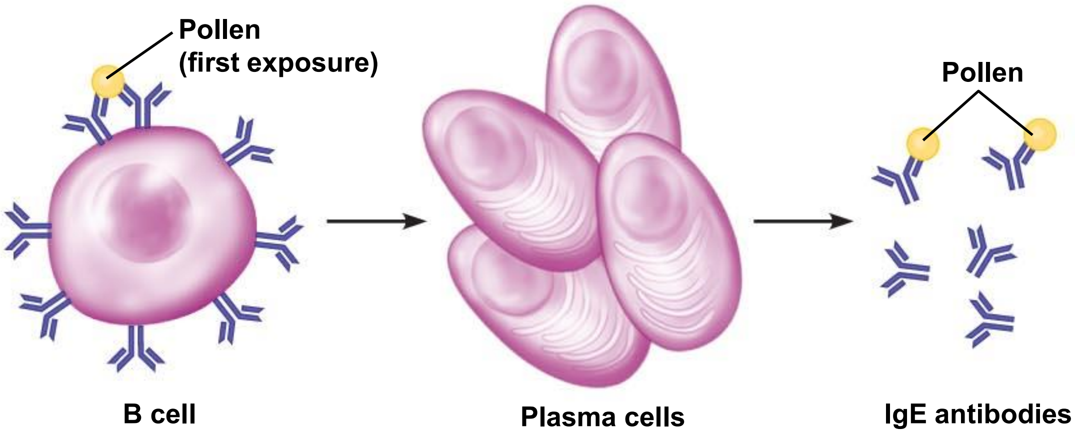
Immune Dysfunctions

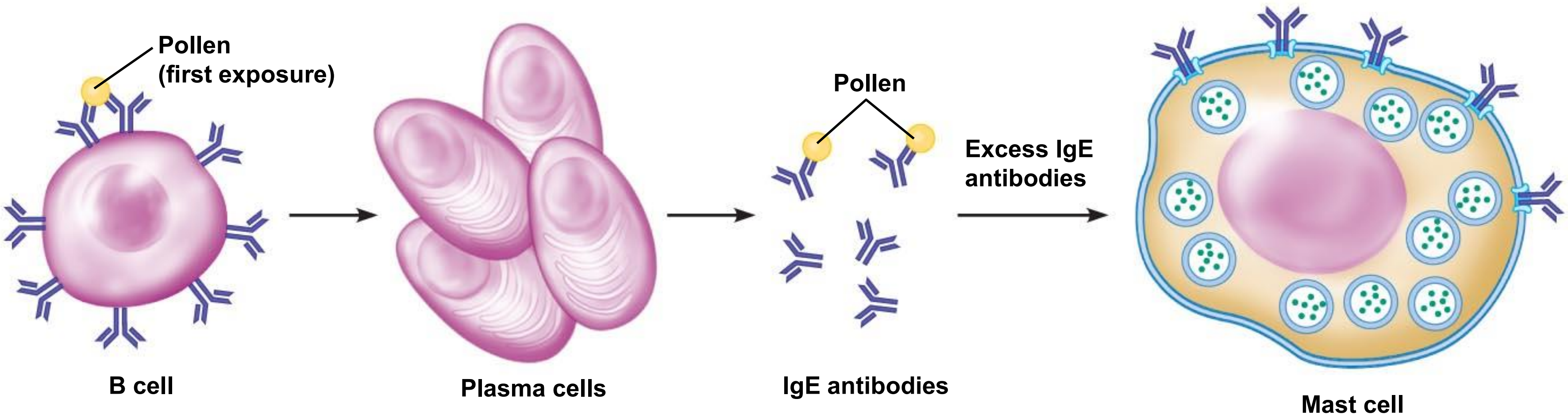
- Allergy: hypersensitive immune response
  - Antigens that induce allergic reactions: allergens
  - Antibodies of allergies: IgE
  - IgE activates mast cells to release histamine

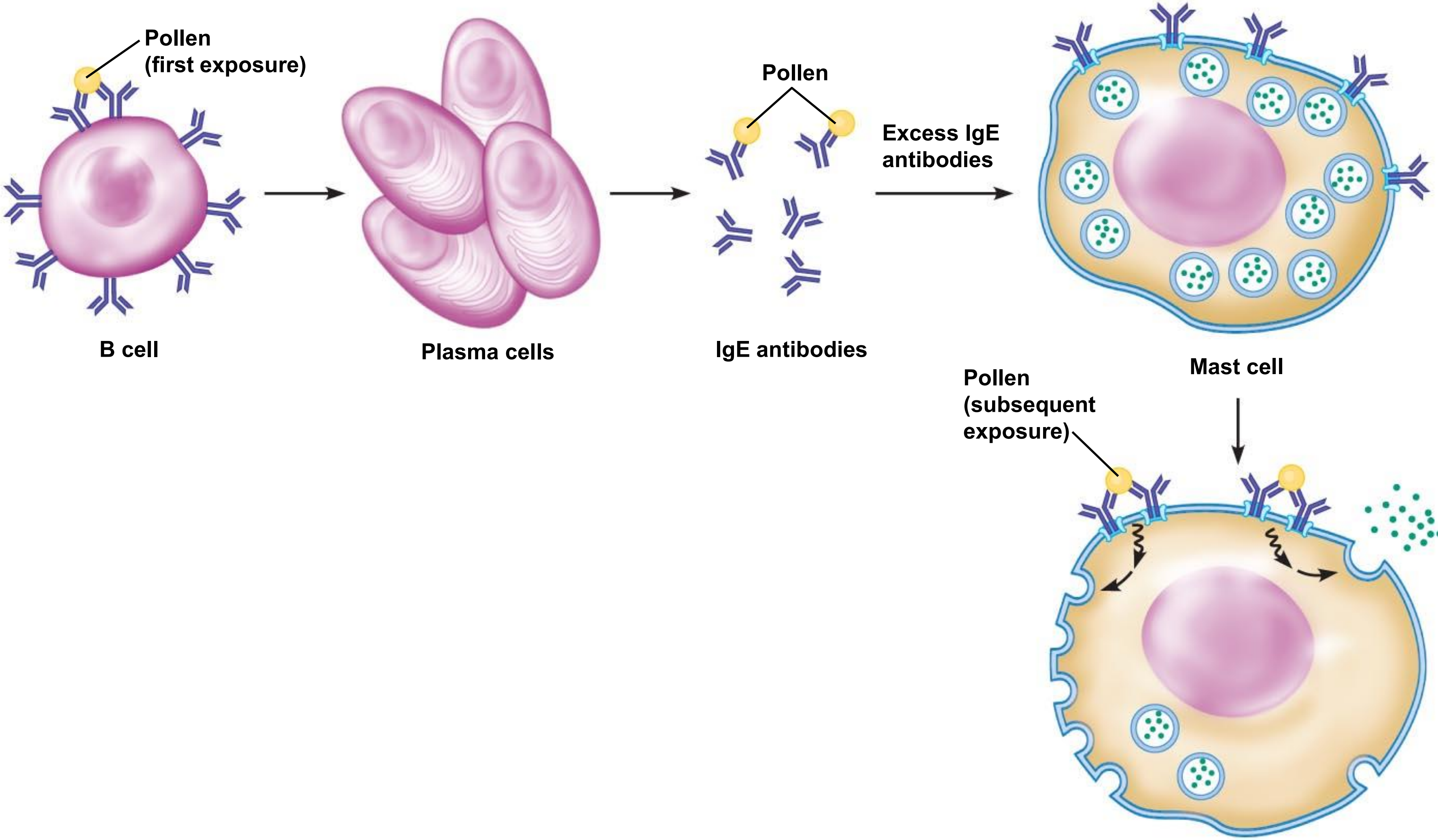

Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies
Excess IgE
antibodies
Mast cell
Pollen
(subsequent
exposure)
Histamine and other
inflammatory agents
Vasodilation
Increased
capillary
permeability
Symptoms of
hay fever:
coughing, sneezing,
sniffles

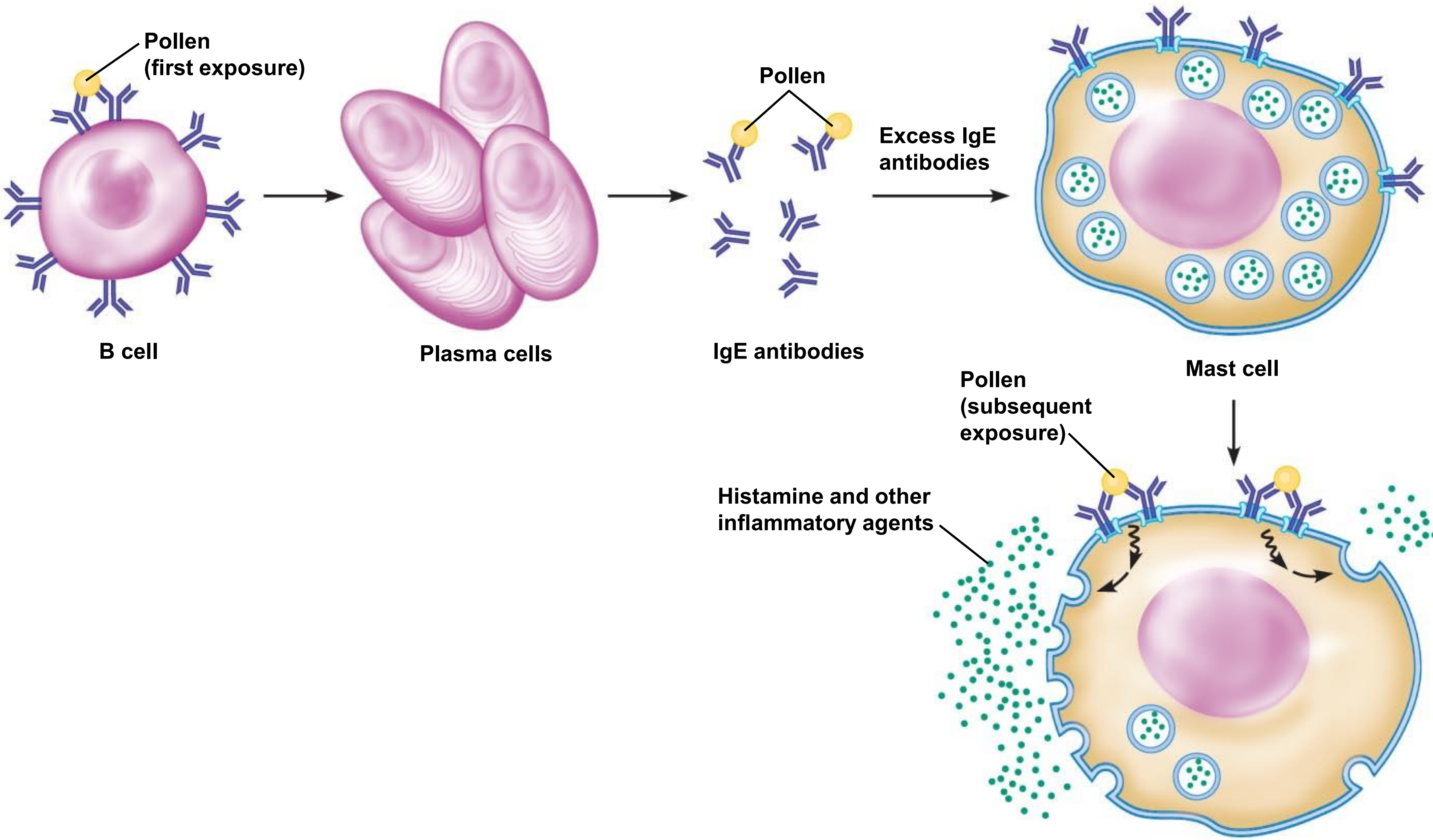

Pollen
(first exposure)
B cell

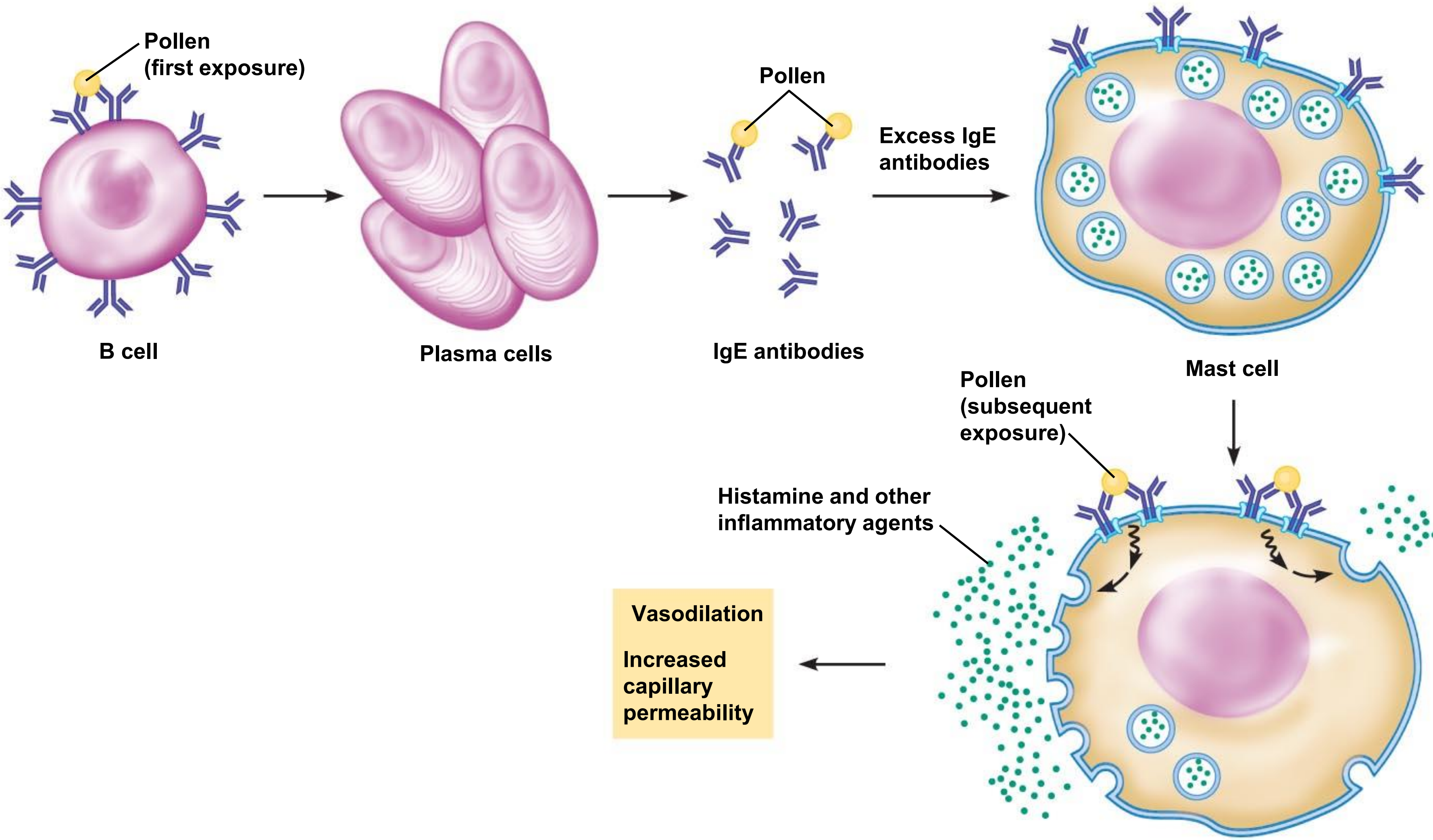

Pollen
(first exposure)
B cell
Plasma cells

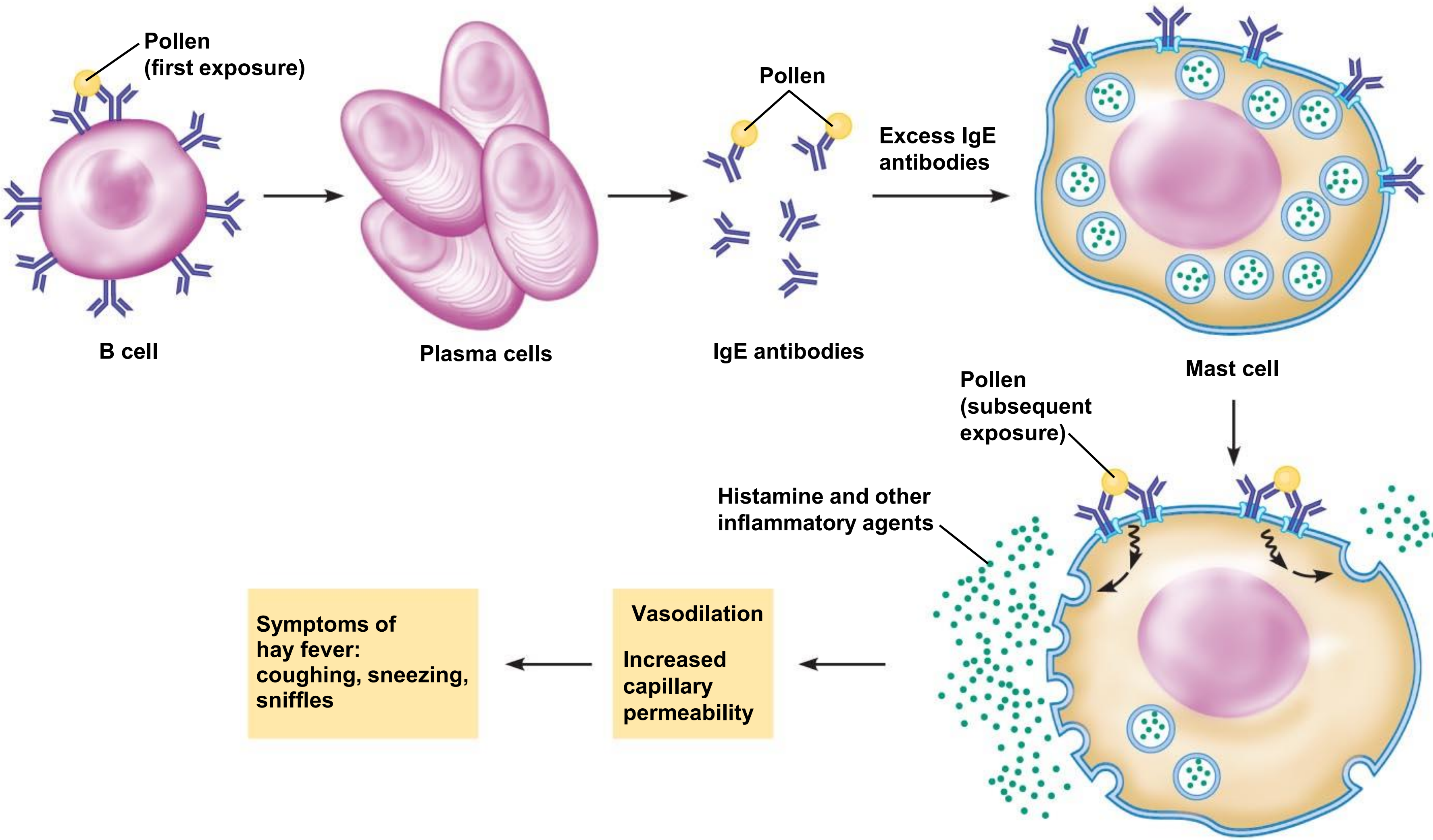

Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies


Pollen
(first exposure)
Pollen
Excess IgE
antibodies
B cell
Plasma cells
IgE antibodies
Mast cell


Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies
Excess IgE
antibodies
Mast cell
Pollen
(subsequent
exposure)


Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies
Excess IgE
antibodies
Mast cell
Pollen
(subsequent
exposure)
Histamine and other
inflammatory agents


Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies
Excess IgE
antibodies
Mast cell
Pollen
(subsequent
exposure)
Histamine and other
inflammatory agents
Vasodilation
Increased
capillary
permeability


Pollen
(first exposure)
B cell
Plasma cells
Pollen
IgE antibodies
Excess IgE
antibodies
Mast cell
Pollen
(subsequent
exposure)
Histamine and other
inflammatory agents
Vasodilation
Increased
capillary
permeability
Symptoms of
hay fever:
coughing, sneezing,
sniffles

Immune Dysfunctions

- Anaphylactic shock
  - Severe allergic reaction
    - Massive release of histamine from mast cells throughout the body
    - Vasodilation → decreased TPR → decreased MAP
  - Prophylactic: epinephrine
    - Increases CO and TPR → increased MAP

Immune Dysfunctions

- Autoimmune diseases
  - Immune system treats a part of self like a pathogen
  - Immune response is induced against that part
  - Examples
    - Insulin-dependent diabetes mellitus
    - Lupus
    - Rheumatoid arthritis
    - Multiple sclerosis

Immune Dysfunctions

- Immunodeficiency diseases
  - Weak or underactive immune system
  - A problem in any factors of immune response can impair immune function
  - Examples
    - Severe combined immunodeficiency disease (SCID)
      - Affects humoral and cell-mediated immunity
    - Hodgkin's disease: cancer of lymphatic system
    - AIDS: affects helper T cells

Immune Dysfunctions

- The role of stress in the immune response
  - Stress suppresses the immune system
  - Steroid hormones (cortisol)
    - Decrease number of leukocytes
    - Anti-inflammatory activity
  - Environmental chemicals
  - Autonomic neural input to lymphoid tissue
  - Multiple levels of interaction between stress and immune system